University of Liberal Arts Bangladesh

Department of Computer Science & Engineering

STA 2101: Statistics & Probability

Lecture 12: Conditional Probabilities

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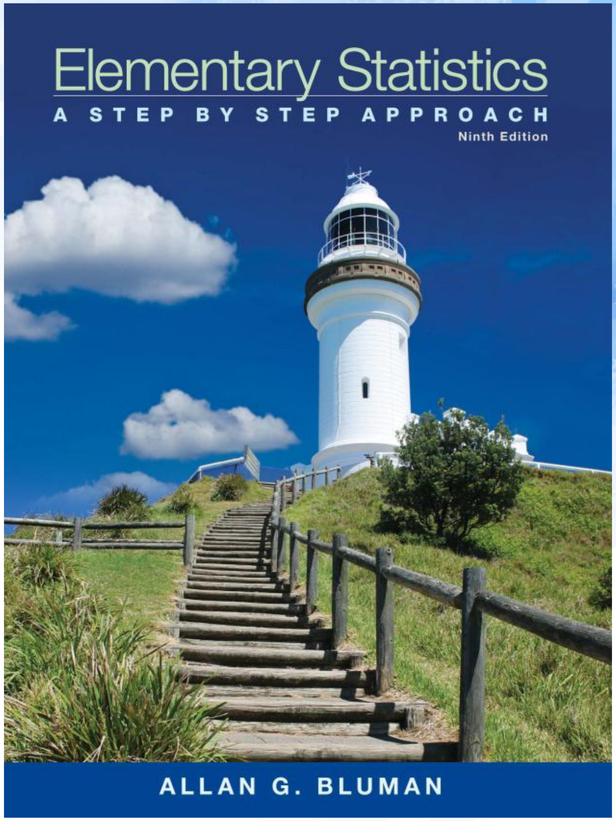
Reference Book

- **Text Book(s)**:
- Elementary Statistics – Allan G. Bluman
- Introduction to Probability and Statistics (14th Edition) By William Mendenhall, Robert J. Beaver, Barbara M. Beaver

- **Reference Material/Book(s)**:
- S Introduction to probability and statistics for engineers and scientists, Sheldon M Ross



Mark Distribution

Assessments	%
Mid Term Examination	25
Final Examination	40
Attendance and Class Participation	5
CT	5
Project	25
Total	100

Today’s Topics

- Conditional Probability

"Probability Of"   "Given"

$$P(\,A \text{ and } B\,) = P(\,A\,) \times P(\,B \mid A\,)$$

Event A   Event B

"Probability of ***event A and event B*** *equals the probability of* ***event A*** *times the probability of* ***event B given event A****"*

EXAMPLE 4–23 Tossing a Coin

A coin is flipped and a die is rolled. Find the probability of getting a head on the coin and a 4 on the die.

SOLUTION

The sample space for the coin is H, T; and for the die it is 1, 2, 3, 4, 5, 6.

$$P(\text{head and }4) = P(\text{head}) \cdot P(4) = \frac{1}{2} \cdot \frac{1}{6} = \frac{1}{12} \approx 0.083$$

The problem in Example 4–23 can also be solved by using the sample space

H1 H2 H3 H4 H5 H6 T1 T2 T3 T4 T5 T6

The solution is $\frac{1}{12}$, since there is only one way to get the head-4 outcome.

EXAMPLE 4–25 Selecting a Colored Ball

An urn contains 3 red balls, 2 blue balls, and 5 white balls. A ball is selected and its color noted. Then it is replaced. A second ball is selected and its color noted. Find the probability of each of these.

a. Selecting 2 blue balls

b. Selecting 1 blue ball and then 1 white ball

c. Selecting 1 red ball and then 1 blue ball

SOLUTION

a. $P(\text{blue and blue}) = P(\text{blue}) \cdot P(\text{blue}) = \frac{2}{10} \cdot \frac{2}{10} = \frac{4}{100} = \frac{1}{25} = 0.04$

b. $P(\text{blue and white}) = P(\text{blue}) \cdot P(\text{white}) = \frac{2}{10} \cdot \frac{5}{10} = \frac{10}{100} = \frac{1}{10} = 0.1$

c. $P(\text{red and blue}) = P(\text{red}) \cdot P(\text{blue}) = \frac{3}{10} \cdot \frac{2}{10} = \frac{6}{100} = \frac{3}{50} = 0.06$

EXAMPLE 4–28 Overqualified Workers

In a recent survey, 33% of the respondents said that they feel that they are overqualified (O) for their present job. Of these, 24% said that they were looking for a new job (J). If a person is selected at random, find the probability that the person feels that he or she is overqualified and is also looking for a new job.

SOLUTION

$$P(\text{O and J}) = P(\text{O}) \cdot P(\text{J}|\text{O}) = (0.33)(0.24) \approx 0.079$$

There is about a 7.9% probability that the person feels that he or she is overqualified and is also looking for a new job.

EXAMPLE 4–29 Homeowner's and Automobile Insurance

World Wide Insurance Company found that 53% of the residents of a city had homeowner's insurance (H) with the company. Of these clients, 27% also had automobile insurance (A) with the company. If a resident is selected at random, find the probability that the resident has both homeowner's and automobile insurance with World Wide Insurance Company.

SOLUTION

$$P(\text{H and A}) = P(\text{H}) \cdot P(\text{A}|\text{H}) = (0.53)(0.27) = 0.1431 \approx 0.143$$

There is about a 14.3% probability that a resident has both homeowner's and automobile insurance with World Wide Insurance Company.

EXAMPLE 4–30 Drawing Cards

Three cards are drawn from an ordinary deck and not replaced. Find the probability of these events.

a. Getting 3 jacks

b. Getting an ace, a king, and a queen in order

c. Getting a club, a spade, and a heart in order

d. Getting 3 clubs

SOLUTION

a. $P(3 \text{ jacks}) = \frac{4}{52} \cdot \frac{3}{51} \cdot \frac{2}{50} = \frac{24}{132{,}600} = \frac{1}{5525} \approx 0.0002$

b. $P(\text{ace and king and queen}) = \frac{4}{52} \cdot \frac{4}{51} \cdot \frac{4}{50} = \frac{64}{132{,}600} = \frac{8}{16{,}575} \approx 0.0005$

c. $P(\text{club and spade and heart}) = \frac{13}{52} \cdot \frac{13}{51} \cdot \frac{13}{50} = \frac{2197}{132{,}600} = \frac{169}{10{,}200} \approx 0.017$

d. $P(3 \text{ clubs}) = \frac{13}{52} \cdot \frac{12}{51} \cdot \frac{11}{50} = \frac{1716}{132{,}600} = \frac{11}{850} \approx 0.013$

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EXAMPLE 4–40 Distribution of Blood Types

There are four blood types, A, B, AB, and O. Blood can also be Rh+ and Rh−. Finally, a blood donor can be classified as either male or female. How many different ways can a donor have his or her blood labeled?

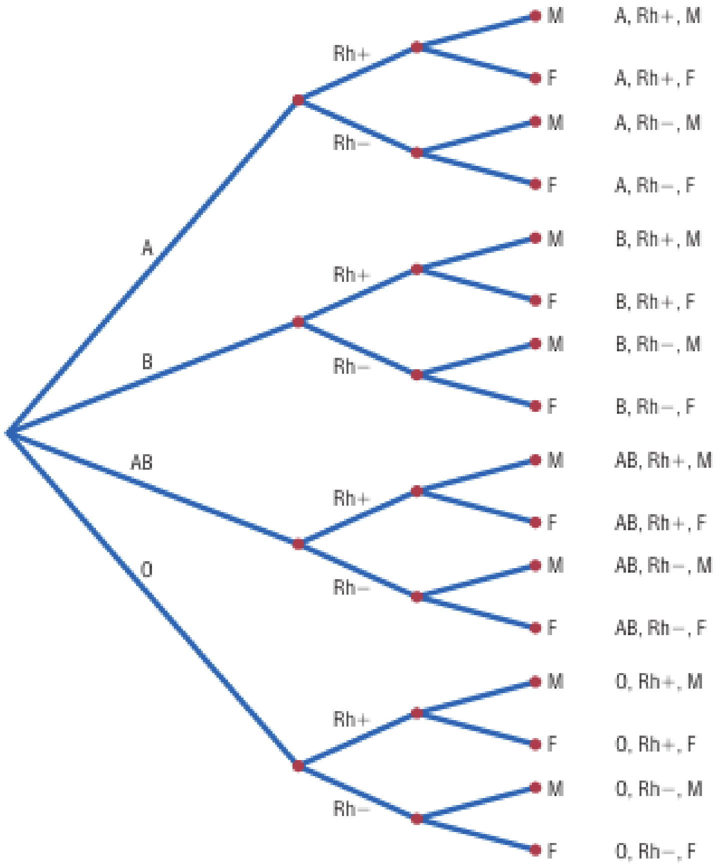
SOLUTION

Since there are 4 possibilities for blood type, 2 possibilities for Rh factor, and 2 possibilities for the gender of the donor, there are $4 \cdot 2 \cdot 2$, or 16, different classification categories, as shown.

Blood type		Rh		Gender	
$\boxed{4}$	·	$\boxed{2}$	·	$\boxed{2}$	= 16

A tree diagram for the events is shown in Figure 4–11.

FIGURE 4–11
Complete Tree Diagram for Example 4–40



Permutations & Combinations

Quick Review

Permutation Rule 1

The arrangement of n objects in a specific order using r objects at a time is called a *permutation of n objects taking r objects at a time*. It is written as ${}_nP_r$, and the formula is

$$ {}_nP_r = \frac{n!}{(n-r)!} $$

The notation ${}_nP_r$ is used for permutations.

$$ {}_6P_4 \text{ means } \quad \frac{6!}{(6-4)!} \quad \text{or} \quad \frac{6!}{2!} = \frac{6 \cdot 5 \cdot 4 \cdot 3 \cdot \cancel{2} \cdot \cancel{1}}{\cancel{2} \cdot \cancel{1}} = 360 $$

Although Examples 4–42 and 4–43 were solved by the multiplication rule, they can now be solved by the permutation rule.

In Example 4–42, five locations were taken and then arranged in order; hence,

$$ {}_5P_5 = \frac{5!}{(5-5)!} = \frac{5!}{0!} = \frac{5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{1} = 120 $$

(Recall that $0! = 1$.)

In Example 4–43, three locations were selected from 5 locations, so $n = 5$ and $r = 3$; hence,

$$ {}_5P_3 = \frac{5!}{(5-3)!} = \frac{5!}{2!} = \frac{5 \cdot 4 \cdot 3 \cdot \cancel{2} \cdot \cancel{1}}{\cancel{2} \cdot \cancel{1}} = 60 $$

EXAMPLE 4–44 Radio Show Guests

A radio talk show host can select 3 of 6 special guests for her program. The order of appearance of the guests is important. How many different ways can this be done?

SOLUTION

Since the order of appearance on the show is important, there are $_6P_3$ ways to select the guests.

$$_6P_3 = \frac{6!}{(6-3)!} = \frac{6!}{3!} = \frac{6 \cdot 5 \cdot 4 \cdot 3!}{3!} = 120$$

Hence, there can be 120 different ways to select 3 guests and present them on the program in a specific order.

EXAMPLE 4–45 School Musical Plays

A school musical director can select 2 musical plays to present next year. One will be presented in the fall, and one will be presented in the spring. If she has 9 to pick from, how many different possibilities are there?

SOLUTION

Order is important since one play can be presented in the fall and the other play in the spring.

$$_9P_2 = \frac{9!}{(9-2)!} = \frac{9!}{7!} = \frac{9 \cdot 8 \cdot 7!}{7!} = 72$$

There are 72 different possibilities.

Permutation Rule 2

The number of permutations of n objects when r_1 objects are identical, r_2 objects are identical, . . . , r_p objects are identical, etc., is

$$\frac{n!}{r_1!\, r_2! \cdots r_p!}$$

where $r_1 + r_2 + \cdots + r_p = n$.

EXAMPLE 4–46 Letter Permutations

How many permutations of the letters can be made from the word *STATISTICS*?

SOLUTION

In the word *STATISTICS*, there are 3 S's, 3 T's, 2 I's, 1 A, and 1 C.

$$\frac{10!}{3!3!2!1!1!} = \frac{10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{3 \cdot 2 \cdot 1 \cdot 3 \cdot 2 \cdot 1 \cdot 2 \cdot 1 \cdot 1 \cdot 1} = 50{,}400$$

There are 50,400 permutations that can be made from the word *STATISTICS*.

Combination Rule

The number of combinations of r objects selected from n objects is denoted by ${}_nC_r$ and is given by the formula

$$ {}_nC_r = \frac{n!}{(n-r)!r!} $$

EXAMPLE 4–47 Combinations

How many combinations of 4 objects are there, taken 2 at a time?

SOLUTION

Since this is a combination problem, the answer is

$$ {}_4C_2 = \frac{4!}{(4-2)!2!} = \frac{4!}{2!2!} = \frac{\overset{2}{\cancel{4}} \cdot 3 \cdot \cancel{2!}}{\cancel{2} \cdot 1 \cdot \cancel{2!}} = 6 $$

This is the same result shown on the previous page.

Notice that the expression for ${}_nC_r$ is

$$ \frac{n!}{(n-r)!r!} $$

which is the formula for permutations with $r!$ in the denominator. In other words,

$$ {}_nC_r = \frac{{}_nP_r}{r!} $$

This $r!$ divides out the duplicates from the number of permutations. For each two letters, there are two permutations but only one combination. Hence, dividing the number of permutations by $r!$ eliminates the duplicates. This result can be verified for other values of n and r. *Note:* ${}_nC_n = 1$.

EXAMPLE 4–48 Flier Types

An advertising executive must select 3 different photographs for an advertising flier. If she has 10 different photographs that can be used, how many ways can she select 3 of them?

SOLUTION

$$_{10}C_3 = \frac{10!}{(10-3)!3!} = \frac{10!}{7!3!} = \frac{10 \cdot 9 \cdot 8}{3 \cdot 2 \cdot 1} = 120$$

Hence, there are 120 different ways to select 3 photographs from 10 photographs.

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EXAMPLE 4–50 Four Aces

Find the probability of getting 4 aces when 5 cards are drawn from an ordinary deck of cards.

SOLUTION

There are ${}_{52}C_5$ ways to draw 5 cards from a deck. There is only 1 way to get 4 aces (that is, ${}_4C_4$), but there are 48 possibilities to get the fifth card. Therefore, there are 48 ways to get 4 aces and 1 other card. Hence,

$$P(4\text{ aces}) = \frac{{}_4C_4 \cdot 48}{{}_{52}C_5} = \frac{1 \cdot 48}{2{,}598{,}960} = \frac{48}{2{,}598{,}960} = \frac{1}{54{,}145}$$

EXAMPLE 4–51 Defective Transistors

A box contains 24 transistors, 4 of which are defective. If 4 are sold at random, find the following probabilities.

a. Exactly 2 are defective.

b. None is defective.

c. All are defective.

d. At least 1 is defective.

SOLUTION

There are ${}_{24}C_4$ ways to sell 4 transistors, so the denominator in each case will be 10,626.

a. Two defective transistors can be selected as ${}_4C_2$ and two nondefective ones as ${}_{20}C_2$. Hence,

$$P(\text{exactly 2 defectives}) = \frac{{}_4C_2 \cdot {}_{20}C_2}{{}_{24}C_4} = \frac{1140}{10{,}626} = \frac{190}{1771}$$

b. The number of ways to choose no defectives is $_{20}C_4$. Hence,

$$P(\text{no defectives}) = \frac{_{20}C_4}{_{24}C_4} = \frac{4845}{10{,}626} = \frac{1615}{3542}$$

c. The number of ways to choose 4 defectives from 4 is $_4C_4$, or 1. Hence,

$$P(\text{all defective}) = \frac{1}{_{24}C_4} = \frac{1}{10{,}626}$$

d. To find the probability of at least 1 defective transistor, find the probability that there are no defective transistors, and then subtract that probability from 1.

$$\begin{aligned} P(\text{at least 1 defective}) &= 1 - P(\text{no defectives}) \\ &= 1 - \frac{_{20}C_4}{_{24}C_4} = 1 - \frac{1615}{3542} = \frac{1927}{3542} \end{aligned}$$

EXAMPLE 4–52 Magazines

A store has 6 *TV Graphic* magazines and 8 *Newstime* magazines on the counter. If two customers purchased a magazine, find the probability that one of each magazine was purchased.

SOLUTION

$$P(1\ \textit{TV Graphic}\text{ and }1\ \textit{Newstime}) = \frac{_6C_1 \cdot {_8C_1}}{_{14}C_2} = \frac{6 \cdot 8}{91} = \frac{48}{91}$$

Thank You

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